

Phosphorus

Test Details

METHODOLOGY

Colorimetric

RESULT TURNAROUND TIME

Within 1 day

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Use

Causes of high phosphorus: Youth; exercise; dehydration and hypovolemia; high phosphorus content enema; acromegaly; hypoparathyroidism; pseudohypoparathyroidism; bone metastases; hypervitaminosis D; sarcoidosis; milk-alkali syndrome; liver disease, such as portal cirrhosis; catastrophic events such as cardiac resuscitation, pulmonary embolism, renal failure; diabetes mellitus with ketosis; serum artifact—sample not refrigerated; overheated, hemolyzed sample, or serum allowed to remain too long on the clot.

Although phosphate accumulation occurs as renal disease progresses, hyperphosphatemia is not a feature of early renal failure;¹ it does not usually develop before renal function has diminished to about 25% of normal.² Osteitis fibrosa in uremic subjects, from excessive bone turnover, relates to hyperphosphatasia. The role of hyperphosphatemia in promotion of such secondary hyperparathyroidism is well established.³ A relationship to osteomalacia in hemodialysis patients exists.³

Causes of low phosphorus: (Hypophosphatemia may occur with or without phosphate depletion. Serum levels vary as much as 2.0 mg/dL during the day.)

Very severely malnourished subjects may have low phosphate levels, but even in starvation, phosphorus levels usually are normal. Antacids, diuretics, and long term steroids are among the common agents bearing a relationship to severe hypophosphatemia.⁴ Recent carbohydrate ingestion decreases phosphorus, as does intravenous glucose administration; cases of hypophosphatemia relate to I.V. carbohydrate,⁴ dialysis, hyperalimentation, prolonged intravenous administration of phosphate-free fluids, metabolic states involving glucose, potassium, and pH. Depletion of phosphate occurs in diabetic ketoacidosis. Like potassium, phosphorus returns to the cell with therapy of diabetic ketoacidosis and serum levels may diminish significantly during treatment. Osmotic diuresis induced by glycosuria in poorly controlled diabetes may lead to urinary phosphate losses with negative phosphorus balance. PO_4 levels may prove useful in initiation of insulin therapy, in diabetic ketoacidosis and other situations of insulin lack; with hyperglucagonemia, corticosteroid and epinephrine use, and in respiratory alkalosis. Association of hypophosphatemia with impaired glucose metabolism is thought to reflect decreased tissue sensitivity to insulin.⁵ Alcoholism and other hepatic disorders are found very frequently among patients with low PO_4 . Alcoholic ketosis and alcohol withdrawal are among causes of hypophosphatemia. There is a slight decrease in serum phosphorus in the last trimester of pregnancy.

Primary hyperparathyroidism and other causes of calcium elevation, including ectopic hyperparathyroidism (pseudohyperparathyroidism).

Patients with sepsis, including Legionnaires' disease and other respiratory infections. Twenty-two percent of instances of respiratory infections had serum phosphorous ≤ 2.4 .⁶ Halevy and Bulvik report gram-negative septicemia as a common cause of severe hypophosphatemia among 55,000 chemistry profiles of hospitalized patients they studied.⁴ (Hypophosphatemia impairs bactericidal activity).

Vitamin D deficiency; osteomalacia, inherited and sporadic forms of hypophosphatemic rickets. In work-up for osteomalacia, look for decreased calcium and phosphorus and increased alkaline phosphatase. Biopsy, however, can be abnormal even when these biochemical parameters are within normal limits.

Renal tubular disorders (Fanconi syndrome, renal tubular acidosis); use of antacids that bind phosphorus (look for hypercalciuria, low urinary phosphorus, high alkaline phosphatase);⁷ dialysis, vomiting; saline or lactate I.V.; steatorrhea, malabsorption, severe diarrhea, nasogastric suction; hypokalemia; negative nitrogen balance; decreased dietary PO_4 intake; recovery from severe burn injury; salicylate poisoning; acute gout; tumor-related: described as including hemangiopericytomas (uncommon pathologic entities) and neurofibromatosis; transfusion of blood; arteriography.

The signs and symptoms of phosphate depletion may include neuromuscular, neuropsychiatric, gastrointestinal, skeletal, and cardiopulmonary systems. Manifestations usually are accompanied by serum levels <1.0 mg/dL.

Severe hypophosphatemia is most common in elderly patients and is often found in postoperative subjects.⁴

Complications of hypophosphatemia: Effect on RBC 2,3-diphosphoglycerate and oxygen dissociation.⁸ Depression of myocardial function (contractibility), decreased cardiac output; respiratory failure and respiratory muscle weakness; increased incidence of sepsis, impairment of bactericidal activities.⁹ CNS consequences: polyradiculopathy, paresthesias, tremor, ataxia, weakness, slurred speech, stupor, coma, seizure; joint stiffness; myopathy; renal stones, hypercalciuria secondary to renal phosphate leak; insulin resistance, glucose intolerance. Rhabdomyolysis may complicate marked hypophosphatemia. A mortality rate of 20% is described in patients whose phosphorus concentration was 1.1–1.5 mg/dL.⁴

Custom Additional Information

Increasing dietary intake of potassium has been reported to increase serum phosphate concentrations apparently by decreasing renal excretion of phosphate.¹⁰ During the last trimester of pregnancy there is a sixfold increase in calcium and phosphorus accumulation as the fetus triples its weight. Plasma phosphorus concentrations may provide a useful means to assess response to phosphate supplements in the premature infant.¹¹

Specimen Requirements

Specimen

Serum (preferred) **or** plasma

Volume

1 mL

Minimum Volume

0.7 mL (**Note:** This volume does **not** allow for repeat testing.)

Container

Red-top tube, gel-barrier tube, **or** green-top (lithium heparin) tube. Do **not** use oxalate, EDTA, or citrate plasma.

Collection Instructions

Separate serum or plasma from cells within 45 minutes of collection.

Stability Requirements

Temperature	Period
Room temperature	7 days
Refrigerated	14 days
Frozen	14 days
Freeze/thaw cycles	Stable x3

Reference Range

See table.

Age	Male* (mg/dL)	Female* (mg/dL)
0 to 6 m	4.2–7.8	3.4–7.9
7 m to 1 y	4.0–6.8	4.0–6.8
2 to 19 y	3.4–5.5	3.3–5.1
>19 y	2.8–4.1	3.0–4.3
*LabCorp internal studies.		

Storage Instructions

Maintain specimen at room temperature.

Causes for Rejection

Hemolysis; gross lipemia; improper labeling; unspun or improperly spun specimen

Footnotes

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10. Sebastian A, Hernández RE, Portale AA, Colman J, Tatsuno J, Morris RC Jr. Dietary potassium influences kidney maintenance of serum phosphorus concentration. *Kidney Int*. 1990 May; 37(5):1341-1349. PubMed 2345430 (<http://www.ncbi.nlm.nih.gov/pubmed/2345430>)
11. Mayne PD, Kovar IZ. Calcium and phosphorus metabolism in the premature infant. *Ann Clin Biochem*. 1991 Mar, 28(Pt 2):131-142. PubMed 1859151 (<http://www.ncbi.nlm.nih.gov/pubmed/1859151>)